

Reader: You mentioned that the existence of the limit of a function $\left(\frac{f(x)}{g(x)}\right)$ at a , when both $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$ (the existence of the limit of the type $\frac{0}{0}$), is very important for further considerations. Why?

Author: The point is that one of the most important concepts in calculus, namely, that of derivative, is based on the limit of the type $\frac{0}{0}$. This will be clear in the subsequent dialogues.
